exploration and use of outer space for the benefit of mankind, encourages private sector use, and ensures tenets of U.S. national security.

The next panelist, *Prof. Stephen Gorove* (USA) of the University of Mississippi Law Center, spoke on the lingering issues of the 1979 Moon Agreement and addressed some of the policy choices for the future. He pointed out that several international organizations have been considering the implications of the establishment of an international lunar base, which could be a logical step following the building of the international space station and a precursor to a Mars mission sometime in the first half of the next century. He sketched the broad contours of the governing legal framework applicable to the moon and other celestial bodies which have been laid down in the 1967 Outer Space Treaty and were reaffirmed and further expanded upon in the Moon Agreement. He drew attention to the fact that the Moon Agreement failed to receive sufficient support from the space-faring nations, largely due to its provisions dealing with the exploitation of lunar resources which were to be regarded as "the common heritage of mankind." In conclusion, he noted that the UN General Assembly, following the recommendation of COPUOS, decided not to consider at this time a review of the Moon Agreement, which under the circumstances appeared to be the correct choice according to Professor Gorove since actual exploitation which could have justified a review remained far off sometime in the 21st century.

Finally, *Prof. Kim* discussed issues of State liability for damage caused by space debris. Although debris poses a danger to activities in outer space, the present international legal framework provides insufficient regulation to prevent increased space debris. He stressed that international cooperative effort must be undertaken to monitor and remedy the situation.

Stephen Gorove
Chair, Ed. Bd., J. SPACE L.

COMMERCIALIZATION AND ECONOMICS OF SPACE OPERATIONS AND APPLICATIONS, SEARCH FOR EXTRA-TERRESTRIAL INTELLIGENCE, AND INTERNATIONAL SPACE PLANS FOR MARS

During the recent Congress of the International Astronautical Federation (IAF), held in Oslo on October 2-6, 1995, the International Academy of Astronautics (IAA) conducted many sessions on a wide variety of topics. Of particular interest were those covering the commercialization and economics of space operations, the legal ramifications involved in the search for extraterrestrial intelligence, and the international space plans for Mars.

Among the papers presented in the session dealing with commercialization of space activities was that of *E. Le Proux de la Riviere*, Euroconsult (France), who discussed the synergy that results when military technology is transferred to the commercial satellite sector. He noted many examples of successful technology transfer, and concluded that Government investment usually works to bring about greater returns in the long-run. He felt the only exceptions to this conclusion would appear to be

Japan, which despite substantial expenditures has only obtained a very small market share in commercial satellites, and developing countries, which upon entering the market have to compete immediately with commercial giants such as Hughes.

P. Willekens of the ESA Space Commercialization Office, *A Poirier* of the Canadian Space Agency, and *H.G. Craft* of Marshall Space Flight Center in Huntsville, Alabama (USA) each elaborated on their countries efforts in promoting technology transfer. Of particular interest was Mr. Craft's presentation, who noted that their method was as simple as creating a toll-free number (1-800-USA-NASA) for any individual or company wishing to inquire about whether NASA possessed the technology to solve their particular problem or create their product idea. The engineers at the Space Center will devote up to 40 hours of their time at no charge in order to answer the query.

J.J. Egan, from the Egan Group/TEG International (USA), challenged the reality of the phrase "space commerce." Beyond the information sector (communications and remote sensing), he felt there was nothing to label 'commercial', since responsibility and control of these activities are for the most part vested in a Government. He warned that if activities in space continue to be colored as if they were commercial simply because of the political benefit involved, then in the long run both government and industry will be hurt.

On the subject of space transportation, his views reflected what appeared to be the general consensus of all speakers, namely:

1) the current launch services market is an oligopoly (small number of suppliers);
2) Prices have remained constant for a long time;
3) There has been some decrease in prices because of competition by Russia and China, but the decrease has been to little to open new areas of space activity such as materials processing and solar power;
4) The transportation cost is "Public enemy #1."

He concluded that the current transportation systems are neither affordable, reliable, nor available. The key to the future appears to be a significant decrease in launching costs. Until that time, 'space commerce' will not become a reality.

In the session pertaining to economics of space applications, of particular interest was the presentation of *R. Obermann*, U.S. Congress, who discussed ways to assess and encourage commercial space initiatives. He noted that the Government's desire to decrease expenditures is at odds with the fact that private participation needs government financial assistance to cover the initial start-up costs. He concluded that "the critical impediment to the establishment of space-based manufacturing capabilities is the inability to have routine, reliable, and affordable access to and from orbit." Although he discussed methods for encouraging private

investment in space manufacturing,* he felt that the negative consequences of that impediment are likely to outweigh any governmental incentives for space-based manufacturing.

The session dealing with the search for extraterrestrial intelligence (SETI) included a presentation by *Amb. A.A. Cocca* (Argentina), who began by describing how the legal system acts as a catalyst of other social sciences, in the sense that law allows many diverse sciences the ability to achieve their goals in a complementary and orderly manner. He noted that this emphasizes the need for a legal framework governing the earthly response after detection of extra-terrestrial life, a situation that would undoubtedly involve numerous sciences in the psychological, technical, and political fields. This was followed by an overview and discussion of the "Declaration of Principles concerning activities following the detection of extra-terrestrial intelligence," which in his view has undoubtedly become a code of conduct for institutions and individuals involved in the search, as has the "Draft Declaration of Principles concerning the sending of communication to extra-terrestrial civilizations."

Dr. Ernst Fasan (Austria) also presented a paper entitled "Legal Consequences of a SETI Detection," which examined the legal results of a verified detection. On the assumption that both species have parallel understandings of our mutual environment and the physical laws which govern it, he noted that among the laws that would naturally evolve between the two species would be the prohibition of damage to the other race, freedom of will, self-defense, and the mutual recognition of living spaces. He felt that these principles have universal applicability, and consequently these areas of mutual understanding will have to be our starting point.

During the session dealing with international space plans and policies, of particular interest was the presentation by *George Morgenthaler*, University of Colorado (USA), whose paper entitled "Report on the IAA Cosmic Study: The International Exploration of Mars" provided a brief summary of the IAA Cosmic study, which he felt reflects the general beliefs of the IAA regarding the benefits of international cooperation in the exploration of Mars. The study included chapters on the importance of Mars exploration, the importance of international cooperation, the possible trajectory paths and architecture of such a mission, and the legal, economic, and human physiological factors involved. His presentation provided a useful overview of future missions to Mars, namely: the U.S. Pathfinder mission in 1996, the Russian MARS 96 mission, the Japanese PLANET-B launch in 1998, and the European INTERMARSNET mission launch in 2003. He also discussed the International Mars Exploration Working Group (IMEWG), which consists of representatives from eleven space agencies and is responsible for developing an international strategy for the exploration of Mars. His presentation concluded by emphasizing the numerous benefits of exploring Mars, the

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* *See, e.g.,* Congressional Notes in this issue of the JOURNAL.